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MASS SPECTROMETRY EXTRACTION AND SELECTION PIPELINE FOR MACHINE LEARNING

Abstract

Systems and methods are provided for obtaining raw mass spectrometry data from samples, determining signals present across the samples, and separating the raw mass spectrometry data into discrete intervals in each of the samples. At each interval of the discrete intervals of the raw mass spectrometry data, a local highest intensity signal, relative to any other signal within each interval, is determined, and a frequency of occurrence of each local highest intensity signal across the samples is determined. A subset of local highest intensity signals is retrieved based on respective frequencies of occurrence of the local highest intensity signals. The subset of the local highest intensity signals is ingested into a machine learning model.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/750,249, filed May 20, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Mass spectrometry separates a solid, liquid, or gaseous sample into individual constituents based on the mass-to-charge ratio of the constituents. Such separation elucidates the composition of a complex sample. Mass spectrometry entails bombarding the sample with an ion source such as an electron beam, which causes the sample to break up into constituents that become positively charged ions. Subsequently, a mass analyzer may separate these constituents according to their mass-to-charge ratios. For example, an electric or magnetic field may be applied to the constituents while the constituents are accelerated. The mass-to-charge ratios may be measured based on amounts of deflection of the constituents. A detector such as an electron multiplier may detect intensities of the constituents at each of different mass-to-charge ratios. A spectrum of intensity as a function of mass-to-charge ratios illustrates intensities, representing amounts of the constituents of the sample, at each of the mass-to-charge ratios. Therefore, mass spectrometry identifies, quantifies, and characterizes the individual constituents of a sample.

[0003] However, implementation of mass spectrometry for analysis of complex biological samples may require coupling to additional chemical approaches for further separating biological components prior to introduction into a mass spectrometer. For example, mass spectrometry may be augmented with upstream chromatography processes, in particular, liquid chromatography (high performance liquid chromatography [HPLC]), that separates a sample, such as bodily fluids, based on chemical properties. Samples may be inputted or injected into a liquid chromatography column, which includes a stationary phase bonded or adsorbed to a surface of the column. Due to differences in binding to the column of individual compounds, molecules, or chemicals with the sample, the individual compounds, molecules, or chemicals are retained within the column for different durations. Thus, liquid chromatography separates the individual compounds, molecules, or chemicals based on their retention times to the column, prior to introduction into a mass spectrometer. An extracted ion chromatogram from a mass spectrometer illustrates intensities, representing amounts of the individual compounds, molecules, or chemicals, sharing the same mass to charge ratio at different retention times. By selecting a particular mass-to-charge ratio, individual compounds, molecules, or chemicals may be separated due to their different retention times.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure, in accordance with one or more various examples, is described in detail with reference to the following figures. The figures are provided for purposes of illustration only and merely depict typical examples.

[0005] FIG. 1 is an exemplary illustration of a computing system that receives raw data from a mass spectrometer, processes, reformats, and/or transforms the raw data in preparation for analysis and other operations by a machine learning model.

[0006] FIG. 2 is an exemplary illustration of a computing component that determines bin values and applies bin values on an extracted ion chromatogram, in a retention time dimension or axis.

[0007] FIG. 3A is an exemplary illustration of a computing component that determines bin values and applies bin values in a mass spectrum, illustrating signal intensities along a mass-to-charge ratio dimension or axis.

[0008] FIG. 3B is an exemplary illustration of a computing component that determines an amount or number of signals detected in a mass spectrum.

[0009] FIGS. 3C-3E are exemplary illustrations of a computing component that applies different bin values in a mass spectrum.

[0010] FIGS. 4A-4C are exemplary illustrations of a computing component that applies different bin values in a mass spectrum, while illustrating a concept of determining frequencies of occurrence of signals across all data samples.

[0011] FIG. 5A is an exemplary illustration of a computing component that generates an image-based representation of mass spectrometry data based on frequencies of occurrence of peak signals in each bin.

[0012] FIG. 5B is an exemplary illustration of a computing component that performs filtering, such as threshold-based filtering, of the image-based representation.

[0013] FIG. 5C is an exemplary illustration of a scenario in which retention times vary among samples and are estimated across the samples.

[0014] FIG. 5D is an exemplary illustration of a relationship between a number of passes or batches through which a subset of signals is extracted and an amount of memory consumed to store the signals per pass or batch.

[0015] FIG. 5E is an exemplary illustration of a computing component that determines or retrieves a mass-to-charge ratio and retention time corresponding to each distinct signal.

[0016] FIG. 5F is an exemplary illustration of a computing component that generates an input for a machine learning model.

[0017] FIGS. 6A-6F are exemplary illustrations of a computing component that expands a window along a retention time axis to capture potentially stray signals. FIGS. 6A-6D illustrate expansion of windows when no conflicts exist between neighboring windows.

[0018] FIGS. 6E-6F illustrate expansion of windows when a conflict exists between neighboring windows.

[0019] FIG. 7 is an exemplary illustration of a computing component that generates an updated input based on an expanded window for a machine learning model.

[0020] FIG. 8 is an exemplary illustration of a computing component that expands a window along a mass-to-charge ratio axis.

[0021] FIG. 9 is an exemplary illustration of a computing component that generates training data for a machine learning model.

[0022] FIGS. 10-12 are exemplary illustrations of a computing component that performs adjustments or normalization of signal intensity across distinct batches or groups.

[0023] FIG. 13 is an exemplary flowchart, illustrating how a computing component eliminates or mitigates noisy mass spectrometry data, while also reducing computing costs of processing mass spectrometry data.

[0024] FIG. 14 is an example computing component that may be used to implement various features of examples described in the present disclosure.

[0025] The figures are not exhaustive and do not limit the present disclosure to the precise form disclosed.

DETAILED DESCRIPTION

[0026] Mass spectrometry, especially when paired with chromatography, has provided a cornucopia of benefits in identification, quantification, and characterization of samples. mass spectrometry may include limitations such as minor errors in measured mass to charge ratios, prevalence of noise, and occasional failure to detect actual signals of compounds, molecules, or chemicals.

Therefore, some actual compounds, molecules, or chemicals present in a sample may be undetected

or difficult to distinguish from noise signals. Moreover, false positives may be included in the raw data from mass spectrometry. Data extraction and processing approaches have not only failed to adequately address such shortcomings, but have also yielded inconsistent results. These limitations are further exacerbated by ever-increasing demands of processing Gargantuan quantities of data, generally at least on a scale of thousands of samples. Generally, the data extraction and processing approaches are ill-equipped to handle such a scale of samples. Moreover, manual processing is infeasible on an order of thousands of samples. Thus, conventional mass spectrometry data extraction techniques are plagued by inefficiency and unreliability.

[0027] Examples described herein address these challenges by implementing an image-based processing approach, rather than a signal-based approach. In particular, a computing component that receives raw data from a mass spectrometer, processes, reformats and/or transforms the raw data, and feeds or inputs the transformed data into a machine learning component or model that is separate from the computing component, or implements a machine learning model that is associated with or within the computing component to analyze the transformed data. Following the implementation of the machine learning model, the computing component, or a separate computing component, may receive the output from the machine learning model. Based on the output, the computing component, or the separate computing component, may perform additional analysis, processing, and/or other functions. For example, the output may include predictions and/or information indicating readings or values of retention time and/or mass to charge ratio across a multitude of samples, along with probabilities of accuracy of such readings or values, or confidence intervals. From such information, the computing component may derive, infer, or determine an elemental or isotopic signature of the sample, and chemical identities or structures of molecules or compounds within the sample. The computing component may, based on such information, perform diagnosis or treatment. In a particular example, if mass spectrometry were performed on blood samples from patients having particular symptoms, raw data from mass spectrometry may be processed and/or transformed by the computing component, then fed into a machine learning model which may output the constituents of the blood sample. From the constituents of the blood sample, the computing component may determine or detect that certain constituents are higher or lower compared to respective levels in non-symptomatic patients or subjects. Thus, the computing component may diagnose one or more particular disease conditions in the symptomatic patients, and/or develop or implement a treatment to restore the levels of the constituents back to normal ranges.

[0028] The examples described herein increases the accuracy of processed mass spectrometry data, by mitigating or eliminating the effects of noise and retaining signals that represent actual constituents of a sample. Additionally, the examples are tailored for a large scale of samples, such as a scale of thousands of samples, thereby attaining both accuracy and efficiency. Therefore, timing and consumption of resources, such as computing resources, are conserved. The examples described herein thus improve the functionality of a computer that carries out processing of mass spectrometry data faster and more accurately, while expediting and increasing reliability and efficacy of further downstream applications such as diagnoses, therapeutics, and prognoses, ultimately resulting in improved quality of life.

[0029] FIG. 1 is an exemplary illustration of computing system **110** including a computing component **111**. The computing component **111** may include one or more hardware processors (e.g., central processing units (CPUs)) and logic **113** that implements instructions to carry out the functions of the computing component **111**, which include, for example, receiving raw data from a mass spectrometer, processing, reformatting, and/or transforming the raw data, and feeding or inputting the transformed data into a machine learning component or model.

[0030] The computing component **111** may include one or more physical devices or servers, or cloud servers on which services or microservices run. The computing component **111** may store, in a database **112**, raw mass spectrometry data from different samples, and/or reformatted, processed,

or transformed mass spectrometry data. In some examples, the computing component **111** may store, at least temporarily, discarded portions of the raw mass spectrometry data, such as portions of the image representation that has been removed or filtered out, as will be illustrated, for example, in FIG. 5B. The database **112** may further store any results generated from the raw mass spectrometry data, such as absolute or relative intensities of signals, or amounts, of individual constituents, and/or respective mass-to-charge ratios and retention times of the constituents. The database **112** may be indexed by an index **115** to categorize or classify the information stored in the database **112**. In some examples, the computing component **111** may cache at least a portion of the information stored in the database **112** in a cache **116**, which may be part of an internal memory structure within the computing component **111**. For example, the computing component **111** may cache any of the data within the database **112** that may be frequently accessed, referenced, or analyzed. For example, if a particular sample is part of different analyses, then information of that sample may be stored in the cache **116**.

[0031] In particular, the computing component **111** may receive raw mass spectrometry data samples **121**, **122**, and **123**, which may be in a data format of a text file and may be converted from a different data format as received from a mass spectrometer. The different data format, in some examples, may be in an extensible Markup Language. The different data format may be base-64 encoded and/or interleaved, and represented as a series of retention time, mass-to-charge ratio, and intensity tuples. Although only three raw mass spectrometry data samples for simplicity, FIG. 1 is not to be construed to mean or imply that the computing component **111** only receives a certain number of raw mass spectrometry data samples at one time instance. The computing component **111** may process any number of raw mass spectrometry data samples, such as on an order of at least a threshold number of samples (e.g., at least thousands of raw mass spectrometry data samples). Any or each of the raw mass spectrometry data samples **121**, **122**, and **123** may be manifested or stored as a tabular representation. However, FIG. 1 illustrates the raw mass spectrometry data samples **121**, **122**, and **123** as a pictorial representation **120** (e.g., a spectral representation), to more clearly illustrate the information that may be encompassed by the raw mass spectrometry data samples **121**, **122**, and **123**. The pictorial representation **120** illustrates that the raw mass spectrometry data samples **121**, **122**, and **123** may include first data **130** generated by liquid chromatography regarding retention time of individual components (e.g., individual compounds, molecules, or chemicals) within the sample, on a first axis, and second data **140** and **141**, corresponding to different retention times, generated by mass spectrometry regarding mass-to-charge ratios of individual constituents within the sample, on a second axis. For example, the first data **130** may include a total ion chromatogram or a base peak chromatogram. Meanwhile, the second data **140** and **141** may include mass spectrograms that indicate mass-to-charge ratios at specific retention times. For example, the second data **140** may correspond to a specific retention time of around 3.9 minutes, at which a local peak is located. The second data **141** may correspond to a specific retention time of around 2.2 minutes, at which another local peak is located.

[0032] In some examples, the first axis and the second axis may be orthogonal. Heights or amplitudes in a h1 direction indicate respective intensities of signals, and/or respective amounts of individual components that correspond to specific retention times. Meanwhile, heights or amplitudes in a h2 direction indicate respective intensities of signals, and/or respective amounts of individual constituents that correspond to specific mass-to-charge ratios.

[0033] Following the receipt of the multiple raw mass spectrometry data samples (hereinafter “data samples”) **121**, **122**, and/or **123**, the computing component **111** may process the multiple data samples. The processing may entail binning, or determining a bin value, in both a retention time axis, as illustrated in FIG. 2, and in a mass-to-charge ratio axis, as illustrated in FIGS. 3A-3E and 4A-4C. FIG. 2 illustrates an extracted ion chromatogram **220**, corresponding to a single data sample. The extracted ion chromatogram **220** includes intensities of signals **261-280** as a function of retention time, at a specific mass-to-charge ratio or a specific range of mass-to-charge ratios.

[0034] Such a procedure of binning may first encompass determining local maxima over different intervals, or bins, of the retention time axis, as illustrated in FIG. 2, and the mass-to-charge ratio axis, as illustrated in FIGS. 3A-3E, at each data sample (e.g., the raw mass spectrometry data sample **121**, the raw mass spectrometry data sample **122**, the raw mass spectrometry data sample **123**, and other data samples). As will be elaborated on subsequently, the local maxima may refer to the highest intensity signal in each interval or bin. In particular, the application of binning may encompass setting or determining a bin value or bin interval (hereinafter “bin value”). A bin value may refer to a particular interval length in which different signals within a particular bin are consolidated or merged into a single signal. Thus, within a single bin, signals originally captured or detected as distinct signals are no longer distinguished, and the computing component **111** may detect only a single signal having a maximum intensity within each bin. For example, referring to FIG. 2, if the computing component **111** determined the bin value in the retention time axis to be 0.125 minutes, then the computing component **111** would detect only a single signal, within a retention time between 0 and 0.125 minutes, a single signal within a retention time between 0.125 minutes and 0.25 minutes, a single signal within a retention time between 0.25 minutes and 0.375 minutes, and so on.

[0035] Increasing a bin value may reduce an amount of data to be processed, thereby decreasing a consumption of time and computing resources. However, a tradeoff of increasing the bin value may be a compromise in an amount of signals captured, or loss of signals. Therefore, the computing component **111** may determine a bin value that addresses both considerations. Generally, the determination of the bin value may be based on an amount of resources, with respect to time and/or computing resources, consumed in processing the data samples, and an amount of signals that would be lost or failed to be processed as a result of applying a particular bin value. In particular, the computing component **111** may determine a number of signals captured across all data samples at different bin values. More specifically, the computing component **111** may determine a bin value such that by increasing the bin value by a particular factor or a particular amount, no signals, or no more than a threshold number or proportion of signals, would be lost or failed to be captured as a result. This principle of determining a bin value may apply along both a retention time axis and a mass-to-charge ratio axis.

[0036] Thus, the computing component **111** may determine a bin value based on an amount or proportion of signals that would be lost or failed to be captured as a result of increasing the bin value. The increase in the bin value may be by discrete factors, for example, by a particular factor such as 2, 5, or 10. In such a manner, the computing component **111** may determine at which bin value the signal loss starts to become unacceptable (e.g., exceed a threshold proportion or threshold amount) upon increasing the bin value by the particular factor. Additionally or alternatively, the computing component **111** may determine a bin value based on an amount or proportion of signals that would be lost or failed to be captured compared to some given bin value.

[0037] In one example, the computing component **111** may set an initial bin value. According to the initial bin value, the computing component **111** may determine a number of captured signals across all the data samples. The computing component **111** may iteratively increase the initial bin value by a factor, and determine, at each iteration, whether an amount of captured signals decreases by more than a threshold proportion compared to a previous iteration. The computing component **111** may determine a particular bin value at which the amount of captured signals decreases by more than a threshold proportion upon increasing the particular bin value by the factor; and determine the particular bin value as the bin value to be applied. In other examples, the computing component **111** may iteratively decrease the initial bin value by a factor, and determine, at each iteration, an increase in an amount of captured signals, if the initial bin value results in an excessive signal loss.

[0038] In particular, the computing component **111** may determine a first total amount of signals captured at the first bin value. In some nonlimiting examples, the first bin value may be 0.01, 0.001, 0.0125, 0.125, 0.03125, or 0.0625 minutes. If the bin value is 0.125 minutes, then bins **201**

having that bin value would be applied. The computing component **111** may further determine a second total number of signals captured at a second bin value, increased or decreased by a factor (e.g., 2, 5, or 10) compared to the particular bin value. For example, the second bin value may be 0.0625 minutes, using bins **211** having that size. If a difference, in number or in proportion, between the second total number of signals and the first total number of signals, or between the second total number of signals and an original total number of signals, is within a threshold, then the amount of signal loss that resulted by increasing the bin value to the second bin value from the first bin value may still be acceptable. In some nonlimiting examples, the threshold may be 1% or 5% with respect to an increase or decrease in the bin value by a factor of two. Then the computing component **111** may determine a third total number signals captured using a third bin value, such as 0.03125 minutes. The computing component **111** may continue to determine an amount of incremental or overall signal loss that resulted by increasing the bin value by a specific factor (e.g., a factor of two). Such a determination may be based on a total amount of signals captured at two consecutive bin values that differ by a factor, or a comparison between a total number of signals at the third bin value and at the first bin value. Once the amount of signal loss exceeds the threshold, then the computing component **111** may determine not to, or refrain from, increasing the bin value to the other bin value. For example, assume that the computing component **111** captured 1000 signals at a bin value of 0.0125 minutes and 970 signals at a bin value of 0.025 minutes, meaning that the signal loss was three percent. However, upon increasing the bin value to 0.05 minutes, the computing component **111** may have captured only 920 signals. The difference between the number of captured signals between the bin values of 0.0125 minutes and 0.05 minutes is eight percent, while the difference between the number of captured signals between the bin values of 0.025 minutes and 0.05 minutes is also over five percent. Thus, no matter what criteria is used to determine the difference of captured signals, the difference would exceed the threshold proportion. The computing component **111** may determine that the bin value is to be 0.025 minutes. The aforementioned procedure is illustrated in more detail in the subsequent FIG. 2. The principles above also apply to determination of the bin value along the mass-to-charge ratio axis, as illustrated in FIGS. 3A-3E and 4A-4C.

[0039] FIG. 2 illustrates application of different bin values in the retention time axis, using the bins **201** having bin values or sizes (hereinafter “bin values”) of 0.125 minutes, the bins **211** having bin values of 0.0625 minutes, and bins **221** having bin values of 0.03125 minutes. The bin values may be indicative of, or analogous to, pixel sizes or pixel resolutions. As previously alluded to, higher bin values entail a higher likelihood of loss of signals because in each bin, only a single signal is selected or extracted. To illustrate a concept of signal loss as a result of increasing a bin value, in FIG. 2, applying a bin value of 0.125 minutes, using the bins **201**, would result in loss of, or failure to capture, at least the signals **262**, **263**, **270**, **272**, **275**, **277**, and **279**. In particular, the signals **262**, **263**, and **264** would all be within a same bin, and the signal **264** has a higher intensity compared to the signals **262** and **263**. Thus, within that bin, only the signal **264** having a highest intensity would be retained. Next, the signals **270** and **269** would both be within a same bin, and the signal **269** has a higher intensity compared to the signal **270**. Thus, within that bin, only the signal **269** would be retained. Next, the signals **272** and **273** would both be within a same bin, and the signal **273** has a higher intensity compared to the signal **272**. Thus, within that bin, only the signal **273** would be retained. Next, the signals **275** and **276** would both be within a same bin, and the signal **276** has a higher intensity compared to the signal **275**. Thus, within that bin, only the signal **276** would be retained. Next, the signals **277** and **278** would both be within a same bin, and the signal **278** has a higher intensity compared to the signal **277**. Thus, within that bin, only the signal **278** would be retained. Overall, applying a bin value of 0.125 minutes would result in a loss of seven out of twenty signals, or 35 percent of the signals.

[0040] Meanwhile, applying a bin value of 0.0625 minutes would result in a loss of the signals **263**, **272**, and **275**. In particular, the signals **263** and **264**, which were previously in the same bin if the

bin value were 0.125 minutes, would now be in different bins after decreasing the bin value from 0.125 minutes to 0.0625 minutes. The signals **262** and **263** would still remain in a common bin, and of those two signals only the signal **262** would be retained because the signal **262** has a higher intensity. The signals **269** and **270**, which were previously in the same bin if the bin value were 0.125 minutes, would now be in different bins after decreasing the bin value from 0.125 minutes to 0.0625 minutes. Next, the signals **272** and **273** would still remain in a common bin, and of those two signals only the signal **273** would be retained because the signal **273** has a higher intensity. Next, the signals **275** and **276** would still remain in a common bin, and of those two signals only the signal **276** would be retained because the signal **276** has a higher intensity. Next, the signals **277** and **278**, which were previously in the same bin if the bin value were 0.125 minutes, would now be in different bins after decreasing the bin value from 0.125 minutes to 0.0625 minutes. Lastly, the signals **279** and **280**, which were previously in the same bin if the bin value were 0.125 minutes, would now be in different bins after decreasing the bin value from 0.125 minutes to 0.0625 minutes. Overall, three out of 20 signals would be lost at a bin value of 0.0625 minutes. [0041] Meanwhile, applying a bin value of 0.03125 minutes would result in a loss of the signal **263**. The signals **262** and **263** would still remain in a common bin, and of those two signals only the signal **262** would be retained because the signal **262** has a higher intensity. The signals **272** and **273** would be separated into different bins as a result of decreasing the bin value from 0.0625 to 0.03125 minutes. The signals **275** and **276** would also be separated into different bins as a result of decreasing the bin value from 0.0625 to 0.03125 minutes. Overall, one out of 20 signals would be lost at a bin value of 0.03125 minutes. By further reducing the bin value to 0.015625 minutes, the signals **262** and **263** may be separated into different bins. In that scenario, doubling the bin value from 0.015625 to 0.03125 minutes would result in an additional, or marginal, loss of signals at a proportion of five percent, or one in twenty signals. If such an additional loss satisfies or falls within a permitted threshold, then the bin size may be determined to be 0.015625 minutes. Otherwise, if such an additional loss fails to satisfy, or falls outside of a permitted threshold, then the bin size may be determined to be 0.0078125 minutes, because by increasing the bin value from 0.0078125 minutes to 0.015625 minutes, no additional signals would be lost. This process described above, as applied to a single data sample, may be repeated for all other data samples. As will be subsequently described with respect to FIGS. 5A-5F, an image-based representation of the data samples (e.g., the data samples **121**, **122**, **123**, and other data samples) may be generated using the determined bin value along the retention time axis, and along the mass-to-charge ratio axis, as will be illustrated in FIGS. 3A-3E and 4A-4C. From the image-based representation, the computing component **111** may then determine frequencies of occurrence of local maxima, in each bin, across all the data samples.

[0042] FIG. 3A illustrates a mass spectrum **320**, which depicts signal intensities as a function of mass-to-charge ratios at a particular retention time. In FIG. 3A, bins **301** have a bin value of 0.1. Meanwhile, bins **331** have a bin value of 0.05; bins **341** have a bin value of 0.025; bins **351** have a bin value of 0.0125. As previously alluded to, only a single signal is selected or extracted within each bin, thereby likely resulting in loss of signals at higher bin values. To illustrate a concept of signal loss as a result of increasing a bin value, in FIG. 3A, a bin value of 0.1 would result in loss of, or failure to capture, at least signals **361-373** because the signals **361-373** are not highest intensity signals within the respective bins, and the computing component **111** obtains or retrieves the local maximum, or the highest intensity signal, in each of the bins. For example, the signal **361** is not a highest intensity signal within the bin between 700.1 and 700.2 because a signal **390** has a higher intensity compared to the signal **361** in that bin. Moreover, the signal **362** is not a highest intensity signal within the bin between 700.2 and 700.3 because a signal **391** has a higher intensity compared to the signal **361** in that bin. Additionally, neither the signal **363** nor the signal **364** is a highest intensity signal within the bin between 700.3 and 700.4 because a signal **392** has a higher intensity compared to the signals **363** and **364** in that bin. Similar reasoning applies to the signals

366-373, which are not highest intensity signals within their respective bins. Therefore, if the computing component **111** were to apply or implement a bin value of 0.1, an excessive or unacceptable amount of signal loss may ensue. Thus, the computing component **111** may apply or implement a bin value that is smaller than 0.1.

[0043] As alluded to previously, with respect to FIG. 2, the computing component **111** may determine a bin value based on an amount or proportion of signals that would be lost or failed to be captured as a result of increasing the bin value, compared to a previous bin value and/or compared to an original number of signals. Alternatively or additionally, the computing component **111** may determine a bin value based on an amount or proportion of signals that would be gained, or additionally captured, as a result of decreasing the bin value, compared to a previous bin value. Any principles described above regarding binning in the retention time axis may also be applicable to binning in the mass-to-charge ratio axis, and vice versa.

[0044] FIG. 3B illustrates signals that would be detected between the mass-to-charge ratios of 700 and 701.1, without binning. The signals include the aforementioned signals **361-373** and the signals **390-392**, and signals **374-387**, which equates to a total of 30 signals. Meanwhile, FIG. 3C illustrates signals that would be detected at a bin value of 0.0125, using the bins **351**. Using a bin value of 0.0125 would still result in detection of all 30 signals previously illustrated in FIG. 3B. FIG. 3D illustrates signals that would still be detected at a bin value of 0.025, using the bins **341**. Using a bin value of 0.0125 would result in loss of signals **375**, **379**, and **371** because of other signals that have higher intensities in the respective bins. In particular, the signal **375** is in a bin between 700.15 and 700.175, and the signal **390** has a higher intensity in that bin. The signal **379** is in a bin from 700.475 to 700.5, and the signal **378** has a higher intensity in that bin. The signal **371** is in a bin from between 700.95 to 700.975, and the signal **385** has a higher intensity in that bin. Thus, changing the bin value to 0.025 would result in a loss of 3 signals, a proportion of ten percent compared to the 30 signals using the bin value of 0.0125. FIG. 3E illustrates signals that would still be detected at a bin value of 0.05 (e.g., using the bins **331**). Using the bins **331** would result in loss of signals **361**, **376**, **377**, **363**, **365**, **366**, **383**, **369**, **372**, and **373** compared to using the bins **341**. Thus, changing the bin value to 0.05 would result in a loss of 10 signals, or a proportion of 10/27 or 37%.

[0045] Using a bin value of 0.05, the signals **361**, **376**, **377**, **363**, **365**, **366**, **383**, **369**, **372**, and **373** would be lost because of other signals that have higher intensities in the respective bins. In particular, the signal **361** is in a bin from 700.15 to 700.2. The signal **390** has a higher intensity in that bin. The signal **376** is in a bin from 700.2 to 700.25.

[0046] The signal **391** has a higher intensity in that bin. The signal **377** is in a bin from 700.25 to 700.3. The signal **362** has a higher intensity in that bin. The signal **363** is in a bin from 700.3 to 700.35. The signal **392** has a higher intensity in that bin. The signal **365** is in a bin from 700.4 to 700.45. The signal **364** has a higher intensity in that bin. The signal **366** is in a bin from 700.6 to 700.65. The signal **381** has a higher intensity in that bin. The signal **383** is in a bin from 700.8 to 700.85. The signal **368** has a higher intensity in that bin. The signal **369** is in a bin from 700.85 to 700.9. The signal **384** has a higher intensity in that bin. The signal **372** is in a bin from 701 to 701.05. The signal **386** has a higher intensity in that bin. The signal **373** is in a bin from 701.05 to 701.1. The signal **387** has a higher intensity in that bin. If the threshold, or permitted loss of signals, is 5%, then the computing component may determine the bin value to be 0.0125, because an increase from the bin value of 0.0125 to 0.025 would result in a 10% loss of signals, which exceeds 5%. If the threshold, or permitted loss of signals, is 10%, then the computing component may determine the bin value to be 0.025, because an increase from the bin value of 0.025 would result in a loss of signals of 10%, which is still within the threshold. In the aforementioned scenarios, the threshold loss of signals corresponds to a difference between numbers of captured signals at two consecutive bin values, differing by some factor, such as 2, 5, or 10. However, the threshold loss of signals may, alternatively, correspond to a difference between a number of

captured signals at a particular bin value and an original number of captured signals, such as illustrated in FIG. 3B.

[0047] Only one mass spectrometry data sample is illustrated in FIG. 2 and the FIGS. 3A-3E. The computing component **111** may implement the aforementioned procedure across all mass spectrometry data samples (e.g., thousands of samples) and determine an overall signal loss resulting from application of different bin values. The overall signal losses, or an overall proportion of signal losses, determined at different bin values may be compared to an overall threshold to determine a particular bin value to be applied across all mass spectrometry data samples. The same determined bin value may be applied across all samples. Although the foregoing focuses on determine a bin size respective to the mass-to-charge ratio axis, the computing component **111** may apply similar or same principles to determine a bin size respective to the retention time axis as well.

[0048] In some examples, when determining the frequencies, the computing component **111** may confirm that the identified local maxima or peaks across different data samples, in a particular bin, correspond to a same signal. Assume that in the bin between 700.225 and 700.25, that a highest intensity signal (e.g., the signal **391**) has an intensity of 2×10^6 . The computing component **111** may then determine frequencies, across other data samples, at which a highest intensity signal within the bin between 700.225 and 700.25 matches or corresponds to the signal **391**. To determine whether an other signal in another data sample matches the signal **391**, the computing component **111** may determine whether the other signal has an intensity within a threshold range of that of the signal **391** (e.g., an intensity of 2×10^6), within that bin. In some nonlimiting examples, the threshold range may be one percent, five percent, ten percent, 0.1% percent, 0.05% percent, or 0.01% percent.

[0049] In some examples, different data samples may have a same signal at slightly different positions or values of mass-to-charge ratios. For example, a same signal may occur at mass-to-charge ratios of 791.5, 791.49999 and 791.49998, which may be in different bins, due to measurement errors of the mass spectrometers, for example. Therefore, when determining frequencies of occurrence, the computing component **111** may expand a window previously bounded by a bin in the retention time axis or a mass-to-charge ratio axis. An amount of expansion may be by a threshold value, range, or proportion, of the mass-to-charge ratio, such as, 0.001, 0.0001, 0.01, or 25×10^{-6} . The computing component **111** may expand a previous window to include the threshold range. For example, if the threshold value is 25×10^{-6} , then a window with a bin value of 0.025, between 791.475 and 791.5, would now be adjusted to be between 791.474975 and 791.500025.

[0050] Additionally, the computing component **111** may determine a reference value of where an actual signal occurs by taking an average, median, or mode over all data samples that have the actual signal present. For example, if the raw mass spectrometry data samples **122** and **123** have the actual signal present at 791.49999 and 791.49998, respectively, and the raw mass spectrometry data sample **121** has the actual signal present at 791.5, the computing component **111** may use an average or median of 791.5, 791.49999, and 791.49998, or 791.49999, as a reference point for the location or position of the actual signal. Using 791.49999 as a reference point, the computing component **111** may determine that any data sample that has a signal, with a proper intensity, corresponding to a mass-to-charge ratio within the threshold range of 791.49999 has the actual signal present. In other words, any data sample that has a signal of a proper intensity within the threshold value of 791.49999, or which deviates by less than the threshold value from 791.49999, may be determined to correspond to the actual signal.

[0051] The computing component **111** may determine and record a particular mass-to-charge ratio and a particular retention time, in each bin. For example, a recorded mass-to-charge ratio, at a particular retention time, may be a mass-to-charge ratio corresponding to a most frequently occurring signal in each mass-to-charge ratio bin. As an illustrative example, the computing component **111** may record the determined mass-to-charge ratio as 700.2332 in the mass-to-charge

ratio bin from 700.225 to 700.25. Determining a most frequently occurring signal may further account for the aforementioned threshold values or ranges with respect to intensities and mass-to-charge ratios or retention times. For example, any signals within a threshold range of intensities, and/or within threshold ranges of mass-to-charge ratios or retention times, may be determined to correspond to the same signal. The recorded mass-to-charge ratios may correspond to an average, median, or mode of all common signals determined to correspond to the most frequently occurring signal. For example, if signals at mass-to-charge ratios of 700.2333, 700.2332, and 700.2331 have all been determined to correspond to the most frequently occurring signal, then the determined mass-to-charge ratio may be 700.2332.

[0052] In some examples, the computing component **111** may compensate for column aging, which may cause shifts in retention time as a mass spectrometry column changes properties over time. In order to correct for retention time drift or shift, the computing component **111** may identify landmark molecules or constituents that are present, or verified to be present, across all samples, and determine retention time shifts with respect to the landmark molecules over time. The determined retention time shifts with respect to the landmark molecules may be applied to other molecules when adjusting for retention time shifts. The mass-to-charge ratios across all samples of the landmark molecules may remain relatively constant, and the landmark molecules may be isolated or segregated from other signals by at least a threshold interval of retention time. That is, no other signals, or no other signals of greater than some threshold intensity, may be present within the threshold interval of retention time from where the landmark molecule is on the retention time axis.

[0053] Upon determining a bin value, the computing component **111** may then convert the data samples (e.g., the data samples **121**, **122**, **123**, and other data samples) into an image format or representation, as illustrated in FIG. 5A, which includes, for each data sample, a single signal in each bin. The image format or representation facilitates further analysis and transformation of the data samples. Each bin, as explained above, may correspond to a given retention time and range of mass-to-charge ratios, or a given mass-to-charge ratio and range of retention times. The computing component **111** may then determine or identify local maxima or peaks in each bin, across all data samples. The computing component **111** may then determine frequencies of occurrence of the local maxima or peaks in each bin across all data samples. For example, if the bin value for mass-to-charge ratio is 0.025, the computing component **111** may determine a single highest intensity signal, or peak (hereinafter “signal”) in a bin between 700 and 700.025, a second single highest intensity signal in a bin between 700.025 and 700.05, a third single highest intensity signal in a bin between 700.05 and 700.075, and so on, for a given data sample. The determination of the highest intensity signal may include determining a particular mass-to-charge ratio and an intensity. The computing component **111** may then determine frequencies, across all data samples, at which respective highest intensity signals occur.

[0054] To further illustrate the concept of determining frequencies, in an example illustration of FIG. 4A, the computing component **111** may obtain multiple mass spectrometry data samples, including a first mass spectrometry data sample **401**, a second mass spectrometry data sample **411**, a third mass spectrometry data sample **421**, and a fourth mass spectrometry data sample **431**. Each of the first mass spectrometry data sample **401**, the second mass spectrometry data sample **411**, the third mass spectrometry data sample **421**, and the fourth mass spectrometry data sample **431** may be implemented as, or similar to, the mass spectrum **320** of any of FIGS. 3A, 3B, 3C, 3D, and 3E. The computing component **111** may determine a total count of signals in each individual bin, across all the aforementioned mass spectrometry data samples. Although FIGS. 4A-4C illustrate mass spectrums, which include data along the mass-to-charge ratio axis, the concepts described are equally applicable to extracted ion chromatograms, as illustrated in FIGS. 3A-3E.

[0055] In FIG. 4A, the computing component **111** may apply bins **451** having a bin value, with respect to a mass-to-charge ratio axis, of 0.05, to each of the aforementioned mass spectrometry

data samples. Using the bins **451**, the computing component **111** determines that in the first mass spectrometry data sample **401**, a signal **402** exists in a bin between mass-to-charge ratios of 700.05 and 700.1, a signal **403** exists in a bin between mass-to-charge ratios of 700.1 and 700.15, a signal **404** exists in a bin between mass-to-charge ratios of 700.15 and 700.2, a signal **405** exists in a bin between mass-to-charge ratios of 700.2 and 700.25, a signal **406** exists in a bin between mass-to-charge ratios of 700.25 and 700.3, a signal **407** exists in a bin between mass-to-charge ratios of 700.3 and 700.35, and a signal **408** exists in a bin between mass-to-charge ratios of 700.35 and 700.4.

[0056] Next, the computing component **111** determines that in the second mass spectrometry data sample **411**, a signal **412** exists in the bin between mass-to-charge ratios of 700 and 700.05, a signal **413** exists in the bin between mass-to-charge ratios of 700.05 and 700.1, a signal **414** exists in the bin between mass-to-charge ratios of 700.1 and 700.15, a signal **415** exists in the bin between mass-to-charge ratios of 700.15 and 700.2, a signal **416** exists in the bin between mass-to-charge ratios of 700.2 and 700.25, a signal **417** exists in the bin between mass-to-charge ratios of 700.25 and 700.3, and a signal **418** exists in the bin between mass-to-charge ratios of 700.3 and 700.35.

[0057] Next, the computing component **111** determines that in the third mass spectrometry data sample **421**, a signal **422** exists in the bin between mass-to-charge ratios of 700 and 700.05, a signal **423** exists in the bin between mass-to-charge ratios of 700.05 and 700.1, a signal **425** exists in the bin between mass-to-charge ratios of 700.15 and 700.2, a signal **426** exists in the bin between mass-to-charge ratios of 700.2 and 700.25, a signal **427** exists in the bin between mass-to-charge ratios of 700.25 and 700.3, and a signal **428** exists in the bin between mass-to-charge ratios of 700.3 and 700.35.

[0058] Next, the computing component **111** determines that in the fourth mass spectrometry data sample **431**, a signal **432** exists in the bin between mass-to-charge ratios of 700 and 700.05, a signal **433** exists in the bin between mass-to-charge ratios of 700.05 and 700.1, a signal **435** exists in the bin between mass-to-charge ratios of 700.15 and 700.2, a signal **436** exists in the bin between mass-to-charge ratios of 700.2 and 700.25, and a signal **437** exists in the bin between mass-to-charge ratios of 700.25 and 700.3.

[0059] The computing component **111** may obtain a sum of occurrences, or frequencies, of signals in each bin across all the samples (e.g., the first mass spectrometry data sample **401**, the second mass spectrometry data sample **411**, the third mass spectrometry data sample **421**, and the fourth mass spectrometry data sample **431**, in addition to other data samples). The computing component **111**, in each bin corresponding to a particular sample, may count at most one signal (e.g., a peak, or highest, intensity signal). In particular, from the four mass spectrometry data samples **401**, **411**, **421**, and **431** illustrated in FIG. 4A, the computing component **111** may determine an existence of a total of three signals in the bin between mass-to-charge ratios of 700 to 700.05, from the signals **412**, **422**, and **432**, a total of four signals in the bin between mass-to-charge ratios of 700.05 to 700.1, from the signals **402**, **413**, **423**, and **433**, a total of two signals in the bin between mass-to-charge ratios of 700.1 to 700.15, from the signals **403** and **414**, a total of four signals in the bin between mass-to-charge ratios of 700.15 to 700.2, from the signals **404**, **415**, **425**, and **435**, a total of four signals in the bin between mass-to-charge ratios of 700.2 to 700.25, from the signals **405**, **416**, **426**, and **436**, a total of four signals in the bin between mass-to-charge ratios of 700.25 to 700.3, from the signals **406**, **417**, **427**, and **437**, a total of three signals in the bin between mass-to-charge ratios of 700.3 to 700.35, from the signals **407**, **418**, and **428**, and a total of one signal in the bin between mass-to-charge ratios of 700.35 to 700.4, from the signal **408**. In some examples, the computing component **111** may determine that within the bin between 700.05 and 700.1, the signal **402** does not correspond to or match the signals **413**, **423**, and **433** due to differences in intensity between the signal **402** and the signals **413**, **423**, and **433**. Thus, even though the signal **402** is a local maximum within the bin between 700.05 and 700.1 for the sample **401**, the signal **402** does not match or correspond to other signals in the same bin between 700.05 and 700.1 for the other

samples **411**, **421**, and **431**. Thus, the signal **402** may not be counted. In some examples, the computing component **111** may determine a frequency of signals that exist across all samples in each bin, as described above, and generate an image representation of such. In such a scenario, the computing component **111** may generate a frequency plot **471**, as shown in FIG. **4B**, illustrating frequencies in each bin as determined above. The frequencies may be illustrated halfway between each bin (e.g., at 700.025 for the bin between 700 and 700.05), at either endpoint of each bin (e.g., at 700 or 700.05), or at any suitable location within each bin.

[0060] In alternative examples, the computing component **111** may additionally determine some statistical measure of the mass-to-charge ratios of the signals that exist. For example, the computing component **111** may determine an average, such as a weighted or overall average, median, or mode, of the mass-to-charge ratios of the samples in each bin. For example, if the signal **412** has a mass-to-charge ratio of 700.01, the signal **421** has a mass-to-charge ratio of 700.02, and the signal **431** has a mass-to-charge ratio of 700.015, then the computing component **111** may determine that an average of the three mass-to-charge ratios would be 700.015. In such a scenario, the computing component **111** may generate a frequency plot **481**, as illustrated in FIG. **4C**, which may illustrate a frequency of three along a y-coordinate and a x-coordinate corresponding to the previously determined average mass-to-charge ratio of 700.015.

[0061] FIG. **5A** illustrates a result of the computing component **111** generating an image-based representation **501**. The image-based representation depicts frequencies or counts of signals across the data samples (e.g., the data samples **121**, **122**, **123**, and other data samples) in each retention time bin and/or mass-to-charge ratio bin. Heights of each of the peaks indicate a frequency or count in which the signals appear across all data samples.

[0062] In FIG. **5A**, signals of particular high frequency appear around a mass-to-charge ratio of 275 and a retention time of 30 seconds, and around a mass-to-charge ratio of 100 and a retention time of 140 seconds, denoted as peaks **510** and **511**, respectively. The computing component **111** may extract a subset of peaks that correspond to a frequency that satisfies a threshold, while discarding or removing a remainder of the signals. The threshold may be defined either in terms of data samples or a proportion of data samples. As merely an illustrative example, extracted peaks by the computing component **111** may include the peaks **510** and **511**, as well as peaks **513**, **514**, **515**, and **516**. As another example, peaks **518** and **519**, which correspond to relatively low frequencies or counts, may be among peaks that are discarded. FIG. **5B** illustrates a filtered image-based representation **502**, in which the peaks **518** and **519** have been filtered out. Only peaks **518** and **519** have been illustrated as filtered out for simplicity; any peaks that fail to satisfy a threshold frequency or count may be filtered out.

[0063] In some examples, a threshold proportion of data samples may be ten percent or a threshold number of samples may be 100. Thus, if one of the peaks indicates that less than ten percent of all data samples have a corresponding signal within a particular bin, meaning that the corresponding signal is absent from over ninety percent of all data samples, then the computing component **111** may remove or filter out that peak and disregard any signals that are actually present in the less than ten percent of all data samples. However, otherwise, if ten percent or more of all data samples have the corresponding signal, then the computing component **111** may retain the peak and the corresponding signal that is present in all data samples. Such a filtering procedure may be a first step in removing noise because if a signal is present in a small proportion or number of samples, such a signal is more likely to constitute noise.

[0064] The computing component **111** may then perform further segmentation, smoothening, filtration, characterization, and/or labelling of the extracted peaks and feed the results into a machine learning component or model (e.g., a machine learning model **590**). The machine learning model may include a neural network classifier or any other supervised or non-supervised machine learning algorithm.

[0065] During a process of segmentation, signals that appear close together, for example, which

have respective mass-to-charge ratios and/or retention times within threshold ranges of one another, may be distinguished. The computing component **111** may distinguish between two signals by inverting the signals and determining whether the two signals have separate falling and rising edges, and/or a demarcation. In particular, as illustrated in FIG. 5B, the peak **516** may be inverted to form an inverted peak **526**. The computing component **111** may determine that the inverted peak **526** includes separate rising and falling edges, such as a first falling edge **530** and a second falling edge **540**, and a first rising edge **531** and a second rising edge **541**. Additionally the computing component **111** may determine a demarcation or boundary **536**. Thus, the computing component **111** may determine that the peak **516** is actually separated into two distinct peaks **546** and **547**. In such a manner, the computing component **111** may distinguish between two separate peaks, or verify an existence of two separate peaks, as in the example of the peak **516**.

[0066] In FIG. 5C, following the identification of peaks, the computing component **111** may perform determination or estimation of retention times over all samples. Due to drift, inherent unique instrument characteristics, and interactions of compounds, retention times may not exactly align across all samples. Therefore, the computing component **111** may obtain an average time at which respective portions of the samples, a compound, or a substance has eluted. This average time may be a weighted centroid or a statistical center, in which half of retention times of the samples are less than the average time and half of retention times of the samples are greater than the average time. A retention time for a single sample may correspond to or be defined by a peak, or local maximum, on an extracted ion chromatogram. To determine retention times across the samples, the computing component **111** may first identify particular bins from the filtered image-based representation **502** that correspond to retention times, at which respective portions of the samples have eluted. For example, the computing component **111** may determine particular retention time bins in which the peak **546** resides by determining positions, along the retention time axis, of the first rising edge **531**, the second rising edge **541**, and the boundary **536**. As an illustrative example, assume that each retention time bin value is 0.001 minutes, and the particular bins identified may be from 0.499 to 0.5, 0.5 to 0.501, and 0.501 to 0.502. The computing component **111**, in FIG. 5C, may determine respective positions of retention time peaks within those bins to be 0.5001, 0.4991, 0.5011, 0.5005, and 0.5015 for samples **580**, **581**, **582**, **583**, and **584**, respectively. The computing component **111** may then determine a median, mean, or mode as the statistical average retention time. If the computing component determines a median, then the retention time would be 0.5005 minutes.

[0067] In some examples, the computing component **111** may, in each mass-to-charge ratio bin, extract or retrieve a subset of the peak intensity signals across all the data samples. These extracted or retrieved samples may be fed, ingested, or inputted into the machine learning model **590**. For example, given a number of data samples, such as 1000 data samples, the computing component **111** may extract peak intensity signals from a portion or proportion thereof, such as 100 data samples or ten percent of the data samples having highest values of peak intensity signals in each mass-to-charge ratio bin. Such an operation, or computation, may involve storage, within the computing component **111** (e.g., within the database **112**, the cache **116**, and/or other computing storage), of the subset of the peak intensity signals, or a representation thereof. Additionally, the computing component **111** may perform further preparation and operations, such as transformation and analysis, on the stored subset of the peak intensity signals. In some examples, the computing component **111** may not have enough computing storage capacity, such as an amount of memory (e.g., random access memory (RAM)) to store the entire subset across an entire mass-to-charge ratio dimension. Therefore, the computing component **111** may determine an available amount of computing storage capacity and subdivide the process of extracting the subset into batches based on the available amount of computing storage capacity. For example, the computing component **111** may reserve a certain proportion, such as 50 percent, of the available amount of computing storage capacity, and determine a corresponding amount of signals that would consume that proportion of

the available amount of computing storage capacity. Thus, if the available amount of computing storage capacity is 100 GB, from which the computing component **111** reserves 50 GB, an amount of signals that consumes 50 GB of storage may be a hundred signals, which may correspond to a mass-to-charge ratio interval of 0.1. The computing component **111** may determine to process each batch in mass-to-charge ratio intervals of 0.1. However, if the available amount of computing storage capacity is 200 GB, the computing component **111** may determine to process each batch in mass-to-charge ratio intervals of 0.2.

[0068] Each batch may correspond to a particular interval of mass-to-charge ratios or a particular interval of retention time and mass to charge ratios. A length of the particular interval may be determined based on the available amount of computing storage capacity. For example, if the entire mass-to-charge ratio axis extends from 700 to 1000, in a first batch, the computing component **111** may extract a first subset of peak intensity signals from all samples within a mass-to-charge ratio interval of 700 to 700.1. In a second batch, the computing component **111** may extract a second subset of peak intensity signals from all samples within a mass-to-charge ratio interval of 700.1 to 700.2. In a third batch, the computing component **111** may extract a third subset of peak intensity signals from all samples within a mass-to-charge ratio interval of 700.2 to 700.3. Such a subdivision addresses the problems of extracting a subset of peak intensity signals from all samples within the entire mass-to-charge ratio axis of 700 to 1000 in a single pass, which may overwhelm the computing storage capabilities of the computing component **111**. As a result, the process may be versatily applied to any scenario of any amount of available computing storage capabilities within a computer, while conserving time by preventing an excessive number of batches.

[0069] To illustrate the problem of extracting from all samples within the entire mass-to-charge ratio axis in a single pass, a total number of signals or peaks, after filtering, may be 1.8 million. Each signal may have a length, such as a number of pixels, of approximately 371. In some examples, each signal may have a length or number of pixels of between 100 and 1000, or between 100 and 500, inclusive. Given 1000 files and 4 bytes to store each unit length of signal, or each pixel, assuming a 32 bit single precision storage, 2.6 terabytes (TB) of data would be needed. If ten percent of the total read data constitutes the subset to be stored, then 0.26 TB of data would be stored. Most computers do not have 0.26 TB of available memory. FIG. 5D illustrates that by increasing a number of batches or passes through the entire mass-to-charge ratio axis, a memory consumed per batch or pass may decrease. For example, if 50 GB of memory is consumed or available, then the computing component **111** may subdivide into ten batches.

[0070] Referring back to FIG. 4A, to illustrate the aforementioned subdivision on a smaller scale, the computing component **111** may extract a first subset of peak intensity signals from all samples within a mass-to-charge ratio interval of 700 to 700.1. Thus, the first subset may include the signals **412**, **422**, and **432** within the bin from 700 to 700.05, if the signals **412**, **422**, and **432** are among the highest intensity peaks within the bin from 700 to 700.05 when compared across signals of all samples. The first subset may also include the signals **413**, **423**, and **433** within the bin from 700.05 to 700.1, if the signals **412**, **422**, and **432** are among the highest intensity peaks within the bin from 700.05 to 700.1 when compared across signals of all samples. Because the signal **402** has a much lower intensity, the signal **402** may not be included within the subset of extracted signals.

[0071] In some examples, the selection of the subset of peak intensity signals may be based not only on respective intensities of the extracted signals (e.g., intensities of peaks), but also based on variances or levels of consistency in respective intensities across different samples, shapes and respective variances or levels of consistency in the shapes across different samples, noise within the signals or surrounding noise of signals across different samples, and/or differences in intensities and shapes of signals between first samples that have a particular compound compared to second samples that are missing the particular compound, or in which the particular compound is not prominent. In some examples, the levels of consistency in the shapes may be determined along different points or locations of the signals, such as along rising or falling edges.

[0072] The computing component **111** may further remove individual signals corresponding to samples that are outliers and/or determined or predicted to be erroneous or defective. In some examples, the computing component **111** may remove any signals in which a sample has a lower than a first threshold intensity and retain any signals in which a median intensity across all samples exceeds a second threshold intensity. Following the selection of the subset of the peak intensity signals, the computing component **111** may obtain, retrieve, or determine the mass-to-charge ratio and the retention times corresponding to the selected or extracted peak intensity signals. In some examples, the computing component **111** may already have determined mass-to-charge ratios and/or retention times of the respective selected or extracted signals corresponding to each of the bins. The computing component **111** may have recorded the mass-to-charge ratios as metadata, as described with respect to FIGS. 3A-3E. For example, referring back to FIGS. 3A-3E, the computing component **111** may have recorded a specific mass-to-charge ratio of 700.2332 in the mass-to-charge ratio bin from 700.225 to 700.25. If already recorded, the computing component **111** may retrieve the specific mass-to-charge ratio and a specific retention time of each bin corresponding to the selected or extracted peak intensity signals.

[0073] Otherwise, if not already recorded, the computing component **111** may determine, via logic, from the selected or extracted signals, a most frequent mass-to-charge ratio and retention time corresponding to each bin, or alternatively, an average, median, or mode of a subset of most frequent mass-to-charge ratios and retention times within particular ranges (e.g., a range of a particular size or magnitude, such as no more than 0.000025, or 25 parts per million). To do so, the computing component **111** may determine, for each sample or for a subset of the samples, a particular mass-to-charge ratio and retention time having a highest value, or local maxima, in each bin. The computing component **111** may then determine highest frequency occurrences of local maxima of the particular mass-to-charge ratio and the particular retention time across all samples. Upon determining the mass-to-charge ratio and the retention time, the computing component **111** may search for occurrences of the local maxima in neighboring bins in order to account for errors or tolerances across the samples. For example, an error in the mass-to-charge ratio dimension may be 25 parts per million.

[0074] As an illustrative example, in FIG. 5E, a first group **550** of datasets includes a first dataset **551**, a second dataset **552**, and a third dataset **553** and a second group **560** of datasets includes a fourth dataset **561**, a fifth dataset **562**, and a sixth dataset **563**. Each of the first dataset **551**, the second dataset **552**, and the third dataset **553** correspond to different samples. Additionally, each of the fourth dataset **561**, the fifth dataset **562**, and the sixth dataset **563** correspond to different samples. The first dataset **551** and the fourth dataset **561** may correspond to a common sample (e.g., a first sample). The second dataset **552** and the fifth dataset **562** may correspond to a common sample (e.g., a second sample). The third dataset **553** and the sixth dataset **563** may correspond to a common dataset (e.g., a third sample). The first group **550** of datasets may be used to determine a mass-to-charge ratio while the second group **560** may be used to determine a retention time. From the first group **550**, the computing component **111** may determine that in the first dataset **551**, a local maximum of mass-to-charge ratio, within a mass-to-charge ratio bin of between 700.225 to 700.25, at a retention time of 99.9875 seconds, occurs at 700.2375. The computing component **111** may determine that in the second dataset **552**, a local maximum of mass-to-charge ratio, within a mass-to-charge ratio bin of between 700.225 to 700.25, also occurs at 700.2375. The computing component **111** may determine that in the third dataset **553**, a local maxima of mass-to-charge ratio, within a mass-to-charge ratio bin of between 700.225 to 700.25, occurs at 700.2375. Therefore, a most frequent occurrence of the local maxima of the mass-to-charge ratio, across the three samples, is at 700.2375, which occurs in two out of three samples, namely, the first sample and the second sample. Meanwhile, the local maximum of the mass-to-charge ratio of 700.235 only occurs in one of the three samples, namely, the third sample.

[0075] The computing component **111** may further determine, or refine a determination, of the

retention time, given a particular mass-to-charge ratio, using the second group **560**. In particular, from the fourth dataset **561**, the computing component **111** may determine that at a fixed mass-to-charge ratio of 700.2375, as determined previously for the first sample, the retention time that corresponds to a highest intensity signal, or local maximum, occurs at 99.9875 seconds. Similarly, from the fifth dataset **562**, the computing component **111** may determine that at a fixed mass-to-charge ratio of 700.2375, as determined previously for the second sample, the retention time that corresponds to a highest intensity signal, or local maximum, occurs at 99.9875 seconds. Next, from the sixth dataset **563**, the computing component **111** may determine that at a fixed mass-to-charge ratio of 700.235, as determined previously for the third sample, the retention time that corresponds to a highest intensity signal, or local maximum, occurs at 99.9 seconds. Therefore, a most frequent occurrence of the local maxima of the retention time, across the three samples, is at 99.9875 seconds, which occurs in two out of three samples, namely, the first sample and the second sample. Meanwhile, the local maximum of the retention time of 99.9 seconds only occurs in one of the three samples, namely, the third sample. Therefore, the computing component **111** may determine that a most frequent occurrence of local maxima is at a retention time of 99.9875 seconds and a mass-to-charge ratio of 700.2375. In some examples, upon such determination, the computing component **111** may retrieve all occurrences of signals that correspond to the determined retention time and the mass-to-charge ratio by searching in bins that include threshold ranges of the retention time and the mass-to-charge ratio. For example, if the error in the mass-to-charge ratio is 25 parts per million, then the mass-to-charge ratio range to account for such error is 700.21 to 700.255. Given a hypothetical bin value of 0.01, then the computing component **111** may search in bins between 700.20 and 700.21, between 700.21 and 700.22, between 700.22 and 700.23, between 700.23 and 700.24, and between 700.24 and 700.25.

[0076] In alternative examples, the computing component **111** may determine a particular range of a particular size or magnitude in which the highest frequency of signals occur, compared to other ranges of a same magnitude or size within a particular bin. In some examples, a size of the ranges may be $0.05 \cdot 10^{-n}$, $0.025 \cdot 10^{-n}$, or $0.01 \cdot 10^{-n}$, wherein n may be an integer between 0 and 4, inclusive. For example, the computing component **111** may determine an average, median, or mode of a subset of mass-to-charge ratios in the range from 700.235 to 700.2375, inclusive. In such a range, a highest frequency of signals may occur compared to other ranges of a size of 0.0025 within the mass-to-charge bin from 700.225 to 700.25. Within the subset, all signals corresponding to the most frequent mass-to-charge ratios may have intensities within threshold ranges of one another (e.g., between 0.95 and 1 times that of a particular intensity). Using the example of FIG. 5E again, the computing component **111** may determine that the local maxima of mass-to-charge ratios occur at 700.2375 in two samples and 700.235 in one sample. Because these local maxima are all within a particular range, the computing component **111** may obtain a weighted average, median, or mode of these local maxima. For example, the weighted average of two occurrences of 700.2375 and one occurrence of 700.235 would be approximately 700.2366667. However, if in one sample, a local maxima of mass-to-charge ratio occurs at 700.2275, such a local maxima may occur outside of the particular range, and may be disregarded during determination of the mass-to-charge ratio.

[0077] In such a manner, the computing component **111** may identify, characterize, and/or label each of the extracted signals prior to inputting into a machine learning model (e.g., the machine learning model **590**). Additionally, the computing component **111** may determine a more accurate value of mass-to-charge ratio, at a higher resolution compared to a range given by the bin value, in order to provide accurate identification of a particular constituent.

[0078] A particular representation of an input into the machine learning model **590** is illustrated in FIG. 5F. The computing component **111** may generate a plot **570** that includes intensities along a z-axis corresponding to, or indicating, respective retention times along a y-axis, for each of different samples along a x-axis. Though not illustrated for simplicity, the plot **570**, or a separate plot, may also include respective mass-to-charge ratios for each of the different samples. The plot **570**, or

information from the plot **570**, may be transformed into an input **575** that represents a top view, from a perspective of view directly above a xy-plane. Although not illustrated for simplicity, the input **575**, or a separate input, may also include respective mass-to-charge ratios for each of the different samples.

[0079] The intensities in the input **575** have been converted to image, or color, representations based on a grayscale spectrum. For example, white may represent a highest normalized intensity, such as a normalized intensity of 1, while black may represent or indicate an absence of a signal, a normalized intensity of 0, or a region outside of a window. In some examples, the computing component **111** may receive an input or indication of a particular window. In other examples, the computing component **111** may determine a particular window within which a certain proportion (e.g., a majority or all of) the signals are situated. In some examples, the particular window may be determined based on a subset of samples, and/or based on segmentation. The particular window may be a region in which signals (e.g., peaks, tops, or maxima of the signals) of the subset of the samples are situated or located. The computing component **111** may determine the most frequent mass-to-charge ratio and retention time corresponding to each bin, as described with respect to FIGS. 5A-5D, or alternatively, an average, median, or mode of a subset of most frequent mass-to-charge ratios and retention times within particular ranges, to determine the particular window. In some examples, the particular window may be determined further based on variabilities (e.g., standard deviations) of mass-to charge ratios and retention times corresponding to each bin, or corresponding to a subset of signals from different samples in each bin. The input **575** may include the particular window, which indicates boundaries within which the machine learning model **590** is confined to analyze. The peak intensities may be normalized so that all values of peak intensities vary between zero and one, prior to being fed into the machine learning model. Once fed into the machine learning model, the machine learning model may infer, predict, or determine a veracity of any signals or potential signals within the particular window, with or without examining outside the particular window.

[0080] Upon determining or receiving the particular window, the computing component **111** may remove windows that span greater than a threshold amount or interval of retention time, such as, an entire time of retention time for a particular experiment. The computing component **111** may further remove or discard retention time windows that fail to satisfy a threshold number of scans, pixels within the image representation, which may signify sizes or intervals of time, such as three scans. In other words, the computing component **111** may further remove or discard retention time windows that are less than a threshold interval of time. The computing component **111** may further remove or discard windows supported by less than a threshold proportion of samples, such as one percent of samples. Thus, if, within a given retention time window, less than the threshold proportion of samples had a signal, then the computing component **111** may remove or discard that given retention time window.

[0081] In some examples, the computing component **111** may expand the particular window, along with other windows, to account for possible stray samples due to retention time shift or drift and/or errors of mass-to-charge ratios. This expansion of windows may occur following selection of a machine learning model (e.g., the machine learning model **590**). The machine learning model may remove a subset (e.g., a portion or all) of windows that lack true signal to mitigate or avoid conflicts that otherwise would occur during window expansion.

[0082] To expand the particular window with respect to retention time, the computing component **111** may obtain shifted, or offset, plots (hereinafter “shifted plots”), and superimpose or overlay the shifted or offset plots as illustrated in FIGS. 6A-6D. In such a manner, the computing component **111** may expand numerous windows simultaneously rather than expanding each window one-by-one. In particular, in FIG. 6A, the computing component **111** may obtain a plot **610**, which may include signals as illustrated in the plot **570** of FIG. 5F, while further including stray signals **614** and **616** and a particular window **612**. The stray signals **614** and **616** may be outside boundaries of

the particular window **612**. The computing component **111** may obtain a first shifted plot **620** after performing a first shift or offset (hereinafter “first shift”) **622** by shifting the plot **610** in a positive y-direction, while maintaining the particular window **612** without shifting. The first shift **622** may have a particular interval, size, or number of pixels, which may be determined based on a variability of a subset of the signals in the plot **610**. Following the first shift **622**, the computing component **111** may overlay, superimpose, or merge an additional region of the first shifted plot **620** that is captured by the particular window **612**, which corresponds to an additional region having a particular interval or size of the first shift **622**. The additional region was not captured by the particular window **612** when applied to the plot **610**. The computing component **111** may disregard any other regions within the particular window **612** of the first shifted plot **620** which have already been captured within the particular window **612** of the plot **610**.

[0083] As illustrated in FIG. **6B**, the computing component **111** may capture a first region **613** of the plot **610**, denoted as d1 and which corresponds to the particular window **612** within the plot **610**. Next, the computing component **111** may capture a second region **623** within the plot **610**, denoted as d2 and which results when the plot **610** is shifted in the positive y-direction by the first shift **622** while maintaining the particular window **612** without shifting. The second region **623** corresponds to the particular window **612** within the first shifted plot **620**. The computing component **111** captures an additional region **633** which is within boundaries of d2 but outside boundaries of d1 while disregarding an other region **643** that is common to, or present in, both d2 and d1. In other words, the other region **643** is within the intersection of d2 and d1. Therefore, the first shift **622** of the particular window **612** may increase a region that originally included d1 to further include a region d2. As a result, when applying the particular window **612** to the first shifted plot **620**, the computing component **111** may capture the stray signal **614**, which was not captured when applying the particular window **612** to the plot **610**.

[0084] As illustrated in FIG. **6A**, the computing component **111** may obtain a second shifted plot **630** after performing a second shift or offset (hereinafter “second shift”) **632** by shifting the plot **610** in a negative y-direction, while maintaining the particular window **612** without shifting. The second shift **632** may have a particular interval, size, or number of pixels, which may be the same interval, size, or number of pixels as the first shift **622** but in an opposite direction as the first shift **622**. Following the second shift **632**, the computing component **111** may overlay, superimpose, or merge a second additional region of the second shifted plot **630** that is captured by the particular window **612**, which corresponds to an additional region having a particular interval or size of the second shift **632**. The additional region was not captured by the particular window **612** when applied to the plot **610**, or to the first shifted plot **620**. The computing component **111** may disregard any other regions within the particular window **612** of the second shifted plot **630** which have already been captured within the particular window **612** of the plot **610**.

[0085] As illustrated in FIG. **6C**, the computing component **111** may capture the first region **613** of the plot **610**, denoted as d1 and which corresponds to the particular window **612** within the plot **610**. Next, the computing component **111** may capture a third region **625** within the plot **610**, denoted as d3 and which results when the plot **610** is shifted in the negative-direction by the second shift **632** while maintaining the particular window **612** without shifting. The third region **625** corresponds to the particular window **612** within the second shifted plot **630**. The computing component **111** captures an additional region **635** which is within boundaries of d3 but outside boundaries of d1 while disregarding an other region **645** that is common to, or present in, both d3 and d1. In other words, the other region **645** is within the intersection of d3 and d1. Therefore, the second shift **632** may increase a captured region that originally included d1 to further include a region d3. Hence, when applying the particular window **612** to the second shifted plot **630**, the computing component **111** may capture the stray signal **616**, which was not captured when applying the particular window **612** to the plot **610**. In summary, by applying both the first shift **622** and the second shift **632** to obtain the first shifted plot **620** and the second shifted plot **630**,

respectively, while subsequently capturing a region within the particular window **612** of the plot **610**, the first shifted plot **620** and the second shifted plot **630**, the computing component **111** superimposes the three aforementioned captured regions to obtain a region that includes **d1**, **d2**, and **d3**, in other words, a union of **d1**, **d2**, and **d3**, as illustrated in FIG. **6D**.

[0086] The above examples illustrated in FIGS. **6A-6D** occur when no conflicts exist between neighboring windows. However, two neighboring windows that are sufficiently close together may be in conflict when both windows are expanded, and the resulting expanded windows at least partially coincide with each other. In such a scenario, the computing component **111** may determine only one of the two neighboring windows to expand based on which window has a higher signal intensity, such as a mean, median, mode, or highest signal intensity. An example of a conflict is illustrated in FIG. **6E**, in which the particular window **612** conflicts with a second particular window **617**. If the particular window **612** is expanded by the first shift **622** and the second particular window **617** is expanded by the second shift **632**, at least respective portions of the resulting expanded windows may coincide. Because each window represents portions of different signals, one signal may not be included in two distinct windows. However, if the computing component **111** expanded both the particular window **612** and the second particular window **617**, one signal may be erroneously included in both the resulting expanded windows. Thus, the computing component **111** may determine which one of the particular window **612** or the second particular window **617** to expand based on which of the particular window **612** or the second particular window **617** has a higher signal intensity. In FIG. **6E**, the signal intensity within the particular window **612** is higher than that within the second particular window **617**. Therefore, the computing component **111** may determine to expand the particular window **612** without expanding the second particular window **617**.

[0087] In FIG. **6E**, a plot **660** may include stray signals **616** and **618**. A first shifted plot **670** illustrates that a stray signal **618** may be captured upon expansion of the second particular window **617** into an expanded window **677**. To obtain the expanded window **677**, the computing component **111** may apply same or similar principles as illustrated in FIGS. **6A-6D**. Meanwhile, a second shifted plot **680** illustrates that the stray signal **616** may be captured upon expansion of the particular window **612** into an expanded window **682**. To obtain the expanded window **682**, the computing component **111** may apply same or similar principles as illustrated in FIGS. **6A-6D**. Because signals within the particular window **612** have higher intensities compared to signals within the second particular window **617**, the computing component **111** may expand the particular window **612** without expanding the second particular window **617**, so that the stray signal **616** is captured but the stray signal **618** may not be captured. In such a manner, the computing component **111** resolves conflicts between two neighboring windows while capturing signals that are likely of higher intensities but disregarding signals that are likely of lower intensities. Therefore, the computing component **111** prioritizes higher intensity signals to preserve fidelity of such signals.

[0088] In other scenarios, if expansion of a first window coincides with a different, unexpanded window, then the computing component **111** may refrain from expanding the first window. For example, in FIG. **6F**, the computing component **111** may determine or receive an indication of a first window **692** and a second window **693**. As illustrated, if the second window **693** were expanded, then a resulting expanded window **694** would partially coincide with the first window **692**. Likewise, if the first window **692** were expanded, then a resulting expanded window **695** would partially coincide with the second window **693**. In such a scenario, the computing component **111** may refrain from expanding both the first window **692** and the second window **693**.

[0089] FIGS. **6A-6F** illustrate a window expansion process with respect to the retention time axis. The window expansion process may expand an original window to redefine boundaries within which the computing component **111** may extract information. Referring back to FIG. **5F**, the computing component **111** may generate an updated input, compared to the input **575**, based on the expanded windows. An updated input **775** having an expanded window is illustrated in FIG. **7**. The

expanded window **775** may capture stray samples **776** and **777**, which were outside of an original, unexpanded window. The computing component **111** may transmit the updated input that includes the expanded window **775** into the machine learning model, which may perform analysis or re-analysis. The computing component **111** may obtain occurrences and/or specific locations of maximum signal intensities within the expanded window. The computing component **111** may determine, within the expanded window, average retention times at which maximum intensities of particular signals occur across all samples to infer or predict the retention times corresponding to particular signals. Assume a simplified illustrative example having three samples, in which a first sample has a local maximum intensity corresponding to a particular signal occurring at 0.75 minutes, a second sample has a local maximum intensity corresponding to the particular signal occurring at 0.755 minutes, and a third sample has a local maximum intensity corresponding to the particular signal occurring at 0.76 minutes. Then, the computing component **111** would infer that the retention time corresponding to the particular signal occurs at 0.755 minutes.

[0090] As explained above, FIGS. **6A-6F** and **7** illustrate the expansion of a retention time window. The computing component **111** may further perform an expansion of a window along the mass-to-charge ratio axis to obtain a range of mass-to-charge ratios and account for an error or tolerance. Such an expansion may be based on the obtained mass-to-charge ratio of the samples, for example, as determined with respect to FIG. **5E**. For example, in FIG. **8**, if an obtained mass-to-charge ratio **810** is 700.2375 and an error is 25 parts per million, a range **812** of the mass-to-charge ratios is between 700.219994 to 700.2550. Given a bin value of 0.025, the range of the mass-to-charge ratios may span three different mass-to-charge ratio bins, a first bin **814** from between 700.2 to 700.225, a second bin **816** from between 700.225 to 700.25 and a third bin **818** from between 700.25 to 700.275. The computing component **111** may extract information from the different mass-to-charge ratio bins **814**, **816**, and **818**. However, if the obtained mass-to-charge ratio is within a proximity of a different mass-to-charge ratio, such that a difference between the obtained mass-to-charge ratio and the different mass-to-charge ratio does not exceed the error or tolerance, then the computing component **111** may not expand a window corresponding to the obtained mass-to-charge ratio, and the different mass-to-charge ratio, along the mass-to-charge ratio axis.

[0091] In such a manner, the computing component **111** leverages an image-based approach to process mass spectrometry data, to extract data that is most likely to represent a true signal within expanded windows while removing or reducing a number of noisy signals, or signals likely to be noise. Signals that are noisy or likely to be noise would probably occur in at most a small proportion of the data samples. Additionally, such an image-based approach further addresses shortcomings of existing signal, or wavelet-based approaches, which assume that mass spectrometry signals have particular shapes. Such an assumption may not always be valid, because mass spectrometry signals may not have Gaussian or symmetric shapes. Therefore, wavelet-based approaches may erroneously determine spurious signals as actual signals and fail to adequately remove noisy signals. In contrast, using an image-based approach, signals that fail to conform to Gaussian or symmetric, shapes may still be detected and not automatically erroneously determined to be noise or spurious.

[0092] The extracted data, with the expanded retention time windows and mass-to-charge ratio windows, may be fed, transmitted, or ingested into the machine learning model (e.g., the machine learning model **590**), which determines or infers existence or absence, or veracity, of signals. As illustrated in FIG. **9**, the machine learning model may require or receive at least a threshold number of true signals and/or at least a threshold number of spurious signals corresponding to each signal in order to determine or infer whether each signal is a true signal. The threshold number of true signals and/or spurious signals may be used to sequentially train the machine learning model. In some examples, a threshold number of true signals may be fed into the machine learning model. If a performance of the machine learning model is unsatisfactory, as determined, for example, by a loss coefficient, a threshold number of spurious signals may be fed into the machine learning

model.

[0093] For example, the threshold number of true signals and/or spurious signals may be one hundred or fifty. As a specific illustrative scenario, if the machine learning model is determining or inferring an existence or absence of a signal at a retention time of 0.73 minutes and a mass-to-charge ratio of 700.025, the machine learning model may obtain a threshold number of true signals at that retention time and that mass-to-charge ratio, or within threshold ranges of that retention time and that mass-to-charge ratio. The threshold number of signals may include a first subset **910** of signals that are expected to be true signals, which may include signals of among highest intensities at that retention time and that mass-to-charge ratio. The threshold number of signals may also include a second subset **920** of signals that are expected to be false or spurious signals, or noise, at that retention time and that mass-to-charge ratio. In such a manner, the machine learning model may distinguish a true signal and a spurious signal at that particular retention time and mass-to-charge ratio. For each input (e.g., the input **775** with expanded mass-to-charge ratio windows), the machine learning model may output an indication or prediction of whether the signal within the expanded retention time window and the expanded mass-to-charge ratio window is true or spurious, and a confidence level or confidence interval of that determination or prediction.

[0094] From the output of the machine learning model, the computing component **111** may perform further quality control. The computing component **111** may retrieve retention times, mass-to-charge ratios, and other metrics or parameters including signal or peak counts across the samples in which each signal is present, corresponding to the signals indicated as true signals by the machine learning model. The computing component **111** may associate or correlate each of the signals indicated as true signals to a specific constituent, molecule, or compound (hereinafter “constituent”) based on their respective mass-to-charge ratios and retention times, and determine whether the specific constituents match with predicted or expected constituents. The computing component **111** may determine a mass-to-charge ratio window and retention time window corresponding to each signal indicated as a true signal as described with respect to FIGS. **6A-6F** and **7**. The computing component **111** may retrieve one or more most frequently occurring signals within each mass-to-charge ratio window and retention time window, and correlate or associate the most frequently occurring signals with respective particular constituents. For example, if a set of samples in a specific experiment is predicted to have glutamate, aspartate, and butyric acid, the computing component **111** may determine whether any of the indicated true signals correlates to glutamate, aspartate, and butyric acid.

[0095] The computing component **111** may merge two signals, which have been indicated as true signals, that are both within an error or tolerance along the mass-to-charge ratio axis and within a threshold retention time of each other, then the two signals may be merged. The merging of the two signals may encompass extracting a higher intensity (e.g., median intensity) signal and/or disregarding a lower intensity signal. In some examples, the error or tolerance may be 10 parts per million, 20 parts per million, or 25 parts per million. In some examples, the threshold retention time may be 0.01 minutes. For example, if a first signal has a mass-to-charge ratio of 700.025, a retention time of 0.73 minutes, and an intensity of 1000, while a second signal has a mass-to-charge ratio of 700.035, a retention time of 0.735 minutes, and an intensity of 500, the computing component **111** may merge the first signal and the second signal by retaining the first signal and discarding or disregarding the second signal.

[0096] The computing component **111** may adjust or normalize (hereinafter “adjust”) intensities to compensate for batch effects or other effects that cause inaccurate or nonuniform intensity readings. The adjusting may occur after merging. For example, the computing component **111** may detect batch effects when different groups or batches of common constituents exhibit a non-randomized distribution of intensities. The distinct batches may correspond to different times, settings, protocols, plates, or other instruments used to run the distinct batches. The computing component **111** may receive an indication of the different batches from experiment run information.

As illustrated in FIG. 10, the computing component 111 may obtain or generate intensities 1010 of a particular constituent (e.g., glutamate) across all samples (e.g., 3050 samples) prior to adjusting of the intensities 1010. The computing component 111 may detect distinct batches 1012, 1014, 1016, 1018, 1020, 1022, 1024, and 1026. In each batch, a median intensity and/or distribution of intensities may have a statistically significant difference from median intensities and/or distributions of intensities in other batches. In some examples, a statistically significant p-value may be 0.01 or 0.001. The respective median intensities are illustrated as dashes within the respective batches 1012, 1014, 1016, 1018, 1020, 1022, 1024, and 1026 in FIG. 10. To adjust the intensities within each of the distinct batches 1012, 1014, 1016, 1018, 1020, 1022, 1024, and 1026, the computing component 111 may divide an intensity at each point, corresponding to a particular sample, by a median intensity specific to the batch to which the point belongs and multiply by a global median intensity 1028 across all samples (e.g., 3050 samples). For example, to adjust an intensity of a point 1013, the computing component 111 may divide the intensity of the point 1013 by a median intensity 932 of the batch 1012 and multiply by the global median intensity 1028. Therefore, all points within the batch 1012 are adjusted downward because the batch 1012 has a higher median intensity 1027 compared to the global median intensity 1027. To adjust an intensity of a point 1025 within the batch 1024, the computing component 111 may divide the intensity of the point 1025 by a median intensity 945 of the batch 1024 and multiply by the global median intensity 1028. More generally, the computing component 111 may obtain adjusted intensities as follows: $A=R \cdot G/B$, wherein A denotes an adjusted intensity at a specific point, R denotes a non-adjusted intensity, G denotes a global median intensity (e.g., 1028) across all samples, and B denotes a batch median intensity (e.g., 1013, 1025). The computing component 111 may repeat this process for all points to obtain adjusted intensities 1060. Other methods of normalization may also be contemplated.

[0097] In FIG. 11, using same or similar principles of adjusting intensities across different batches as illustrated in FIG. 10, the computing component 111 may adjust intensities 1110 across batches 1112-1134 to obtain adjusted intensities 1160. In FIG. 12, using same or similar principles of adjusting intensities across different batches as illustrated in FIG. 9, the computing component 111 may adjust intensities 1110 across batches 1212-1234 to obtain adjusted intensities 1260.

[0098] In some examples, the computing component 111 may determine median intensity value corresponding to positively identified signals. For example, if the machine learning model positively indicates a presence of a signal at a retention time of 0.73 minutes and a mass-to-charge ratio of 700.025, the computing component 111 may determine the median intensity of the peak at that retention time and mass-to-charge ratio, following the quality control and adjusting procedures described above. If the median intensity is less than a specified threshold, the computing component 111 may refrain, or determine not to, further analyze the peak, but retain the information of such peaks. The information may be retained in the database 116.

[0099] The computing component 111 may further detect whether any signal intensities exhibit a non-random trend, such as, decreasing or increasing over time. For example, if any signal intensities of a particular constituent exhibit a decreasing or an increasing trend with respect to a run order (e.g., an order in which samples are injected into the liquid chromatograph mass spectrometer), the computing component 111 may attribute the decreasing or increasing intensities over time to inherent instabilities of particular constituents, rather than differences in original intensities or levels of the particular constituents in samples that were randomized before run. The computing component 111 may compare a rate of decrease or increase over time to a dissociation constant or other measure of degradation or instability of the particular constituent to determine or verify whether the decrease or increase over time is attributed to an inherent property of the particular constituent. For example, creatinine may degrade over time. Thus, even if an original level or concentration of creatinine in a particular sample was constant, samples that are run, injected, or inputted later may exhibit lower intensities of creatinine compared to samples that are

run, injected, or inputted earlier. Additionally, some constituents may increase in level or concentration because those constituents may be formed due to degradation of other constituents. [0100] FIG. 13 illustrates a computing component 1300 that includes one or more hardware processors 1302 and machine-readable storage media 1304 storing a set of machine-readable/machine-executable instructions that, when executed, cause the hardware processor(s) 1302 to perform an illustrative method of selectively retaining or extracting a subset of mass spectrometry data to be fed or ingested into a machine learning model. It should be appreciated that there can be additional, fewer, or alternative steps performed in similar or alternative orders, or in parallel, within the scope of the various examples discussed herein unless otherwise stated. In some examples, steps or instructions (hereinafter “steps”) 1306-1312 may serve as or form part of logic 113 of the computing component 111. The computing component 1300 may be implemented as the computing component 111 of FIGS. 1, 2, 3A-3E, 4A-4C, 5A-5F, 6A-6F, and 7-12. The machine-readable storage media 1304 may include suitable machine-readable storage media described in FIG. 14. FIG. 13 summarizes and further elaborates on some aspects previously described.

[0101] At step 1306, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to obtain raw mass spectrometry data from samples. For example, the raw mass spectrometry data may include first data with respect to retention time in a first axis and second data with respect to a mass-to-charge ratio in a second axis, as illustrated in FIG. 1. The raw mass spectrometry data may be obtained over a threshold number of samples, such as thousands of samples, and in each sample, the raw mass spectrometry data may be in tabular format, with a first column indication retention times, a second column indicating mass-to-charge ratios, and a third column indicating signal intensities. The pictorial representation 120 has been illustrated in FIG. 1, in order to elucidate the particular information that may be encompassed within the raw mass spectrometry data. At step 1308, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to determine signals present across the samples. For example, the determination of signals may encompass determining particular mass-to-charge ratios and retention times corresponding to the signals. At step 1310, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to separate the raw mass spectrometry data into discrete intervals, or bins, in each of the samples. For example, the process of separating into discrete intervals may include determining a bin value in each of the retention time axis and the mass-to-charge ratio axis, as illustrated in FIGS. 2, 3A-3E, and 4A-4C. The bin value in the retention time axis may be uniformly applied to all of the samples. The bin value in the mass-to-charge ratio axis may also be uniformly applied to all of the samples.

[0102] At step 1312, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to determine a local highest intensity signal at each interval of the discrete intervals of the raw mass spectrometry data, relative to any other signal within that interval, as illustrated, for example, in FIGS. 2 and 3A-3E. At step 1314, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to determine, at each interval of the discrete intervals of the raw mass spectrometry data, a frequency of occurrence of each local highest intensity signal across the samples. At step 1316, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to retrieve a subset of local highest intensity signals based on respective frequencies of occurrence of the local highest intensity signals. For example, the subset of the local highest intensity signals may include those local highest intensity signals that have at least a threshold frequency of occurrence, as illustrated in FIG. 5B. At step 1318, the hardware processor(s) 1302 may execute machine-readable/machine-executable instructions stored in the machine-readable storage media 1304 to ingest the subset of the local highest intensity signals into a machine

learning model (e.g., the machine learning model 590). Therefore, the hardware processor(s) 1302 may implement an image-based approach rather than a signal-based approach to eliminate or reduce noise from raw mass spectrometry data. As a result, the data fed into the machine learning model will likely generate more reliable and valuable results.

[0103] FIG. 14 depicts a block diagram of an example computer system 1400 in which various of the examples described herein may be implemented. In some examples, the computer system 1400 may include a cloud-based or remote computing system. For example, the computer system 1400 may include a cluster of machines orchestrated as a parallel processing infrastructure. The computer system 1400 includes a bus 1402 or other communication mechanism for communicating information, one or more hardware processors 1404 coupled with bus 1402 for processing information. Hardware processor(s) 1404 may be, for example, one or more general purpose microprocessors. In some examples, the hardware processor(s) 1404 may implement the logic 113 of the computing component 111, as illustrated in any of FIGS. 1, 2, 3A-3E, 4A-4C, 5A-5F, 6A-6F, and 7-12.

[0104] The computer system 1400 also includes a main memory 1406, such as a random access memory (RAM), cache and/or other dynamic storage devices, coupled to bus 1402 for storing information and instructions to be executed by processor 1404. Main memory 1406 also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the hardware processor(s) 1404. Such instructions, when stored in storage media accessible to the hardware processor(s) 1404, render computer system 1400 into a special-purpose machine that is customized to perform the operations specified in the instructions.

[0105] The computer system 1400 further includes a read only memory (ROM) 1408 or other static storage device coupled to bus 1402 for storing static information and instructions for the hardware processor(s) 1404. A storage device 1410, such as a magnetic disk, optical disk, or USB thumb drive (Flash drive), etc., is provided and coupled to bus 1402 for storing information and instructions.

[0106] The computer system 1400 may be coupled via bus 1402 to a display 1412, such as a liquid crystal display (LCD) (or touch screen), for displaying information to a computer user. An input device 1414, including alphanumeric and other keys, is coupled to bus 1402 for communicating information and command selections to the hardware processor(s) 1404. Another type of user input device is cursor control 1416, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to the hardware processor(s) 1404 and for controlling cursor movement on display 1412. In some examples, the same direction information and command selections as cursor control may be implemented via receiving touches on a touch screen without a cursor.

[0107] The computing system 1400 may include a user interface module to implement a GUI that may be stored in a mass storage device as executable software codes that are executed by the computing device(s). This and other modules may include, by way of example, components, such as software components, object-oriented software components, class components and task components, processes, functions, attributes, procedures, subroutines, segments of program code, drivers, firmware, microcode, circuitry, data, databases, data structures, tables, arrays, and variables.

[0108] In general, the word “component,” “system,” “component,” “database,” “data store,” and the like, as used herein, can refer to logic embodied in hardware or firmware, or to a collection of software instructions, possibly having entry and exit points, written in a programming language, such as, for example, Java, C or C++. A software component may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language such as, for example, BASIC, Perl, or Python. It will be appreciated that software components may be callable from other components or from themselves, and/or may be invoked in response to detected events or interrupts. Software components configured for execution

on computing devices may be provided on a computer readable medium, such as a compact disc, digital video disc, flash drive, magnetic disc, or any other tangible medium, or as a digital download (and may be originally stored in a compressed or installable format that requires installation, decompression or decryption prior to execution). Such software code may be stored, partially or fully, on a memory device of the executing computing device, for execution by the computing device. Software instructions may be embedded in firmware, such as an EPROM. It will be further appreciated that hardware components may be comprised of connected logic units, such as gates and flip-flops, and/or may be comprised of programmable units, such as programmable gate arrays or processors.

[0109] The computer system **1400** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic which in combination with the computer system causes or programs computer system **1400** to be a special-purpose machine. According to one example, the techniques herein are performed by computer system **1400** in response to the hardware processor(s) **1404** executing one or more sequences of one or more instructions contained in main memory **1406**. Such instructions may be read into main memory **1406** from another storage medium, such as storage device **1410**. Execution of the sequences of instructions contained in main memory **1406** causes the hardware processor(s) **1404** to perform the process steps described herein. In alternative examples, hard-wired circuitry may be used in place of or in combination with software instructions.

[0110] The term “non-transitory media,” and similar terms, as used herein refers to any media that store data and/or instructions that cause a machine to operate in a specific fashion. Such non-transitory media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage device **1410**. Volatile media includes dynamic memory, such as main memory **1406**. Common forms of non-transitory media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge, and networked versions of the same.

[0111] Non-transitory media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between non-transitory media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **1402**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

[0112] The computer system **1400** also includes a communication interface **1418** coupled to bus **1402**. Network interface **1418** provides a two-way data communication coupling to one or more network links that are connected to one or more local networks. For example, communication interface **1418** may be an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, network interface **1418** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN (or WAN component to communicated with a WAN). Wireless links may also be implemented. In any such implementation, network interface **1418** sends and receives electrical, electromagnetic or optical signals that carry digital data streams representing various types of information.

[0113] A network link typically provides data communication through one or more networks to other data devices. For example, a network link may provide a connection through local network to a host computer or to data equipment operated by an Internet Service Provider (ISP). The ISP in turn provides data communication services through the world wide packet data communication network now commonly referred to as the “Internet.” Local network and Internet both use electrical, electromagnetic or optical signals that carry digital data streams. The signals through the various networks and the signals on network link and through communication interface **1418**,

which carry the digital data to and from computer system **1400**, are example forms of transmission media.

[0114] The computer system **1400** can send messages and receive data, including program code, through the network(s), network link and communication interface **1418**. In the Internet example, a server might transmit a requested code for an application program through the Internet, the ISP, the local network and the communication interface **1418**.

[0115] The received code may be executed by the hardware processor(s) **1404** as it is received, and/or stored in storage device **1410**, or other non-volatile storage for later execution.

[0116] Each of the processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code components executed by one or more computer systems or computer processors comprising computer hardware. The one or more computer systems or computer processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The various features and processes described above may be used independently of one another, or may be combined in various ways. Different combinations and sub-combinations are intended to fall within the scope of this disclosure, and certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate, or may be performed in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed example examples. The performance of certain of the operations or processes may be distributed among computer systems or computers processors, not only residing within a single machine, but deployed across a number of machines.

[0117] As used herein, a circuit might be implemented utilizing any form of hardware, software, or a combination thereof. For example, one or more processors, controllers, ASICs, PLAS, PALs, CPLDs, FPGAs, logical components, software routines or other mechanisms might be implemented to make up a circuit. In implementation, the various circuits described herein might be implemented as discrete circuits or the functions and features described can be shared in part or in total among one or more circuits. Even though various features or elements of functionality may be individually described or claimed as separate circuits, these features and functionality can be shared among one or more common circuits, and such description shall not require or imply that separate circuits are required to implement such features or functionality. Where a circuit is implemented in whole or in part using software, such software can be implemented to operate with a computing or processing system capable of carrying out the functionality described with respect thereto, such as computer system **1400**.

[0118] As used herein, the term “or” may be construed in either an inclusive or exclusive sense. Moreover, the description of resources, operations, or structures in the singular shall not be read to exclude the plural. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements and/or steps.

[0119] Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. Adjectives such as “conventional,” “traditional,” “normal,” “standard,” “known,” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that may be available or known now or at any time in the future. The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent.

[0120] Unless the context requires otherwise, throughout the present specification and claims, the word “comprise” and variations thereof, such as, “comprises” and “comprising” are to be construed in an open, inclusive sense, that is as “including, but not limited to.” Recitation of numeric ranges of values throughout the specification is intended to serve as a shorthand notation of referring individually to each separate value falling within the range inclusive of the values defining the range, and each separate value is incorporated in the specification as it were individually recited herein. Additionally, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. The phrases “at least one of,” “at least one selected from the group of,” or “at least one selected from the group consisting of,” and the like are to be interpreted in the disjunctive (e.g., not to be interpreted as at least one of A and at least one of B).

Claims

1. A computer-implemented method, comprising: obtaining raw mass spectrometry data from samples; determining signals present across the samples; separating the raw mass spectrometry data into discrete intervals in each of the samples; at each interval of the discrete intervals of the raw mass spectrometry data: determining a local highest intensity signal, relative to any other signal within each interval; and determining a frequency of occurrence of each local highest intensity signal across the samples; retrieving a subset of local highest intensity signals based on respective frequencies of occurrence of the local highest intensity signals; normalizing the subset of the local highest intensity signals, wherein the normalizing comprises: segmenting the subset of the local highest intensity signals; generating a two-dimensional representation identifying normalized peak intensities within windows corresponding to the segmented subset of local highest intensity signals across different samples, wherein each of the windows is based on a size of each of the discrete intervals, wherein two dimensions of the two-dimensional representation comprise a mass-to-charge ratio or a retention time, and a sample number; determining veracities of each of the subset of the local highest intensity signals based on the two-dimensional representation; and based on the determined veracities, inferring one or more constituents of the samples.
2. The computer-implemented method of claim 1, wherein the raw mass spectrometry data comprises retention times, mass-to-charge ratios, and signal intensities of respective assayed molecules; and the determination of the bin value comprises a first bin value with respect to the retention times and a second bin value with respect to the mass-to-charge ratios.
3. The computer-implemented method of claim 2, further comprising generating an image-based representation of the raw mass spectrometry data, wherein the image-based representation indicates the frequencies of occurrence.
4. The computer-implemented method of claim 1, wherein a respective normalized peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time is identified based on a color or shading.
5. The computer-implemented method of claim 4, further comprising removing local highest intensity signals of which the frequencies of occurrence fail to satisfy a threshold frequency.
6. The computer-implemented method of claim 1, wherein the retrieved subset of the local highest intensity signals satisfy a threshold frequency.
7. The computer-implemented method of claim 1, wherein generating the two-dimensional representation comprises generating a three-dimensional representation and transforming the three-dimensional representation into the two-dimensional representation, wherein the two-dimensional representation removes a dimension that identifies a respective peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time.
8. A computing system comprising: one or more processors; and a memory storing instructions that, when executed by the one or more processors, cause the one or more processors to perform: obtaining raw mass spectrometry data from samples; determining signals present across the

samples; separating the raw mass spectrometry data into discrete intervals in each of the samples; at each interval of the discrete intervals of the raw mass spectrometry data: determining a local highest intensity signal, relative to any other signal within each interval; and determining a frequency of occurrence of each local highest intensity signal across the samples; retrieving a subset of local highest intensity signals based on respective frequencies of occurrence of the local highest intensity signals; normalizing the subset of the local highest intensity signals, wherein the normalizing comprises: segmenting the subset of the local highest intensity signals; generating a two-dimensional representation indicating peak intensities within windows corresponding to the segmented subset of local highest intensity signals across different samples, wherein each of the windows is based on a size of each of the discrete intervals, wherein two dimensions of the two-dimensional representation comprise a mass-to-charge ratio or a retention time, and a sample number, and a respective peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time is indicated based on a color or shading; determining veracities of each of the subset of the local highest intensity signals based on the two-dimensional representation; and based on the determined veracities, inferring one or more constituents of the samples.

9. The computing system of claim 8, wherein the raw mass spectrometry data comprises retention times, mass-to-charge ratios, and signal intensities of respective assayed molecules; and the determination of the bin value comprises a first bin value with respect to the retention times and a second bin value with respect to the mass-to-charge ratios.

10. The computing system of claim 9, wherein the instructions further cause the one or more processors to generate an image-based representation of the raw mass spectrometry data, wherein the image-based representation indicates the frequencies of occurrence.

11. The computing system of claim 8, wherein a respective normalized peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time is identified based on a color or shading.

12. The computing system of claim 11, wherein the instructions further cause the one or more processors to perform removing local highest intensity signals of which the frequencies of occurrence fail to satisfy a threshold frequency.

13. The computing system of claim 8, wherein the retrieved subset of the local highest intensity signals satisfy a threshold frequency.

14. The computing system of claim 8, wherein generating the two-dimensional representation comprises generating a three-dimensional representation and transforming the three-dimensional representation into the two-dimensional representation, wherein the two-dimensional representation removes a dimension that identifies a respective peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time.

15. A non-transitory storage medium storing instructions that, when executed by at least one processor of a computing system, cause the computing system to perform a method comprising: obtaining raw mass spectrometry data from samples; determining signals present across the samples; separating the raw mass spectrometry data into discrete intervals in each of the samples; at each interval of the discrete intervals of the raw mass spectrometry data: determining a local highest intensity signal, relative to any other signal within each interval; and determining a frequency of occurrence of each local highest intensity signal across the samples; retrieving a subset of local highest intensity signals based on respective frequencies of occurrence of the local highest intensity signals; normalizing the subset of the local highest intensity signals, wherein the normalizing comprises: segmenting the subset of the local highest intensity signals; generating a two-dimensional representation identifying normalized peak intensities within windows corresponding to the segmented subset of local highest intensity signals across different samples, wherein each of the windows is based on a size of each of the discrete intervals, wherein two dimensions of the two-dimensional representation comprise a mass-to-charge ratio or a retention

time, and a sample number; determining veracities of each of the subset of the local highest intensity signals based on the two-dimensional representation; and based on the determined veracities, inferring one or more constituents of the samples.

16. The non-transitory medium of claim 15, wherein the raw mass spectrometry data comprises retention times, mass-to-charge ratios, and signal intensities of respective assayed molecules; and the determination of the bin value comprises a first bin value with respect to the retention times and a second bin value with respect to the mass-to-charge ratios.

17. The non-transitory medium of claim 16, wherein the instructions further cause the computing system to perform generating an image-based representation of the raw mass spectrometry data, wherein the image-based representation indicates the frequencies of occurrence.

18. The non-transitory medium of claim 15, wherein a respective normalized peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time is identified based on a color or shading.

19. The non-transitory medium of claim 18, wherein the instructions further cause the computing system to perform removing local highest intensity signals of which the frequencies of occurrence fail to satisfy a threshold frequency.

20. The non-transitory medium of claim 15, wherein generating the two-dimensional representation comprises generating a three-dimensional representation and transforming the three-dimensional representation into the two-dimensional representation, wherein the two-dimensional representation removes a dimension that identifies a respective peak intensity corresponding to the sample number and the mass-to-charge ratio or the retention time.
